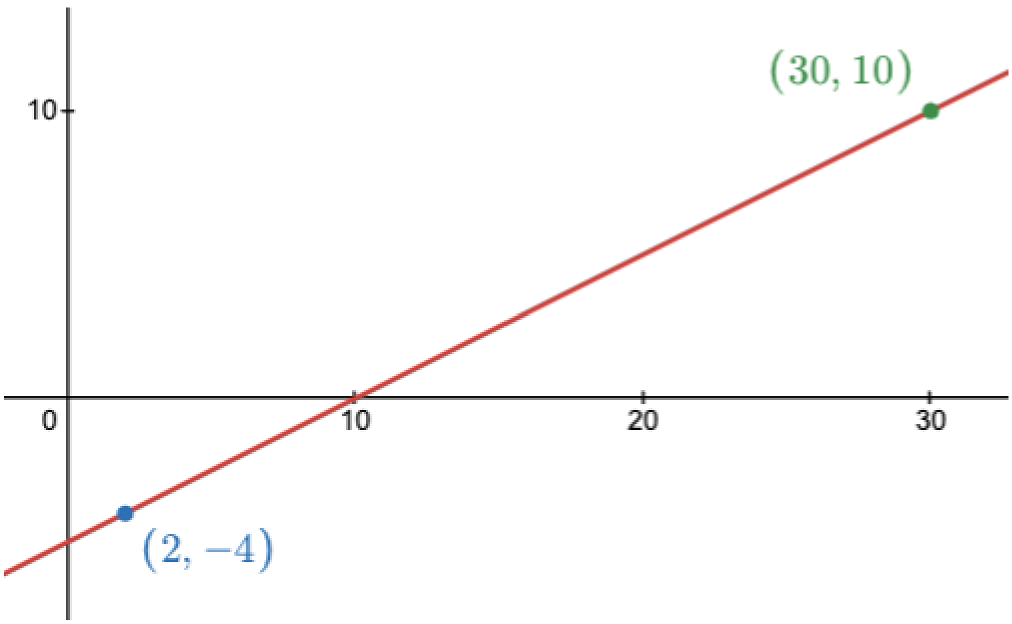
MTH1W

A Formula for Slope

What's the **slope** of this line?



If you are given **any two points** on the line, you can calculate the slope with this formula:

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$ ← the subscripts are NOT numbers, they disappear when you plug numbers in.

(See examples)

Example

1. What is the slope of a line that passes through (10,8) and (-2,32) ?

2. What is the slope of a line that passes through (0,6) and (2,0) ?

3. What is the slope of the line represented by this table of values?

x	y
10	-7
15	-14
20	-21
25	-28

4. A plumber charges a flat fee to show up (unknown amount) and also charges an hourly rate (also unknown).

   A 2-hour job costs $300
   A 5-hour job costs $510

How much does this plumber charge per hour?

Homework

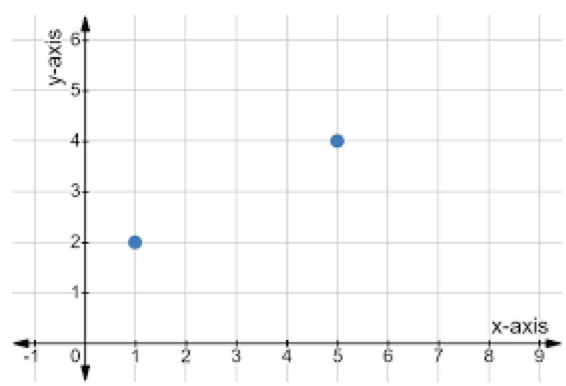
1. Calculate the slope between the two points $(1, -19)$ and $(-2, -7)$

2. Calculate the slope between $(19, -2)$ and $(-11, 10)$ [give answer as reduced, improper fraction]

3. Calculate the slope between $(12, -18)$ and $(-15, -18)$

4. Calculate the slope between $(6, -12)$ and $(15, -3)$

5. Find the slope of this line with "rise over run", AND verify with our new formula.



6. Here is a triangle made from three points: R, S and T.

a) What are the coordinates of R?

b) What are the coordinates of S?

c) What are the coordinates of T?

d) What is the **slope** between R and S?

e) What is the **slope** between R and T?

f) What is the **slope** between S and T?